Minimal block II: uniform envelope and scale limit of an equal-exponent block

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Abstract

For an equal-exponent block $q_0 = \cdots = q_d = p$ with constant $\xi = a$ the product-density formula of unit 73 yields two statements needed for the multiplicity-m state-density theorem: a uniform envelope $c^p Z_{d+1}(c) \le (\prod k_i^{-1}) \frac{1}{d!} E(\beta, L, p, d)\,(1 + \log_+(cB))^d$ valid at every scale $c > 0$ and uniformly in $|a| \le L$, and the scale limit $N^p Z_{d+1}(N\gamma)/\log^d N \to (\prod k_i^{-1}) \frac{1}{d!} \gamma^{-p} A_{p-1}(a)$ for every fixed $\gamma > 0$, by dominated convergence. The tower on any admissible base is also bounded by $K(1 + \log_+ x)^m$ at every $x > 0$. Zero `sorry`/`axiom`.

1 The things formalised

```
theorem iterDivPrim_le_polyLog (hG : LogBase G A M C   ) (m :  ) :
     K :  , 0   K    x, 0 < x → iterDivPrim G m x   K * (1 + logPlus x) ^ m
noncomputable def blockEnv (  L p :  ) (d :  ) :   :=
  Real.exp (  * L ^ 2 / 2)
    *   t in Ioi (0 :  ), (1 + |Real.log t|) ^ d * (t ^ (p - 1) * quadKernel (  / 2) 0 t)
theorem iterChartGen_block_le ... (ha : |a|   L) (c :  ) (hc : 0 < c) :
    c ^ p * iterChartGen   a b k h (d + 1) c
       (  i   Finset.range (d + 1), (1 :  ) / k i) * (1 / (d.factorial :  ))
       * blockEnv   L p d * (1 + logPlus (c * b ^ (  i   Finset.range (d + 1), k i))) ^ d
theorem iterChartGen_block_tendsto ... (h  : 0 <  ) :
    Tendsto (fun N :   => N ^ p / Real.log N ^ d * iterChartGen   a b k h (d + 1) (N *  ))
      atTop (  ((  i   Finset.range (d + 1), (1 :  ) / k i) * (1 / (d.factorial :  ))
       *   ^ (-p) * weightedMass   a (p - 1)))
```

2 Proof strategy

Envelope: on $(0, cB]$, $0 \le \log(cB) - \log t \le \log_+(cB) + |\log t| \le (1 + \log_+(cB))(1 + |\log t|)$ and $e^{-\beta t^2 + \beta a t} \le e^{\beta L^2/2} e^{-(\beta/2)t^2}$; the integral over $(0, cB]$ is then bounded by the integral over $(0, \infty)$ of the nonnegative dominating function (`setIntegral_mono_on`, `setIntegral_mono_set`). Scale limit: after the exact formula and $(N\gamma)^{-p} = N^{-p}\gamma^{-p}$, the integrand is $\mathbf{1}_{(0,N\gamma B]}\big((\log N + \log(\gamma B) - \log t)/\log N\big)^d t^{p-1} e^{-\beta t^2 + \beta a t}$, which converges pointwise to $t^{p-1} e^{-\beta t^2 + \beta a t}$ and is dominated for $\log N \ge 1$ by $(1 + |\log(\gamma B)| + |\log t|)^d t^{p-1} e^{-\beta t^2 + \beta a t}$, integrable by the binomial expansion of unit 73. The tower envelope combines `iterDivPrim_props` on $(0, 1]$ with the complete expansion `iterDivPrim_expansion` on $[1, \infty)$.

3 Roadblocks and resolutions

- `have := hG. _pos` shadows an earlier anonymous `this`; name the facts.
- `positivity` cannot see through `logPlus` or `quadKernel`; supply `logPlus_nonneg`/`quadKernel_pos` explicitly and use `mul_le_mul_of_nonneg_*` instead of `gcongr` when the side goals involve them.
- Rewriting with `mul_assoc` to expose a factor also reassociates the other side; state the inner inequality with `suffices` and finish by `ring`.
- The final identity with $N^p N^{-p} = 1$ closes by `linear_combination` against that fact, after `set`-folding the integral and the product into atoms.

4 Outcome

Astra target 2 is done. Next: the frozen coefficient theorem with noncritical coordinates (dominated convergence over v with the envelope), then the strip estimate for a block with one non-minimal coordinate and the continuity freezing argument.